

Environment

Global Climate Change

Raven

Global Climate Change



Nordic Bulk Carriers

Overview of Chapter 20

- Introduction to Climate Change
 - Causes, Pollutants, Modeling, and Unpredictable Climate Change
- Effects of Climate Change
 - Melting Ice and Rising Sea Level
 - Changes in Precipitation Patterns
 - Effects on Organisms
 - Effects on Human Health
- Effects of Climate Change
 - Effects on Agriculture
 - International Implications
 - Links
- Dealing with Global Climate Change
 - Mitigation
 - Adaptation
 - International Efforts to Reduce Emissions

Reduced Arctic Sea Ice

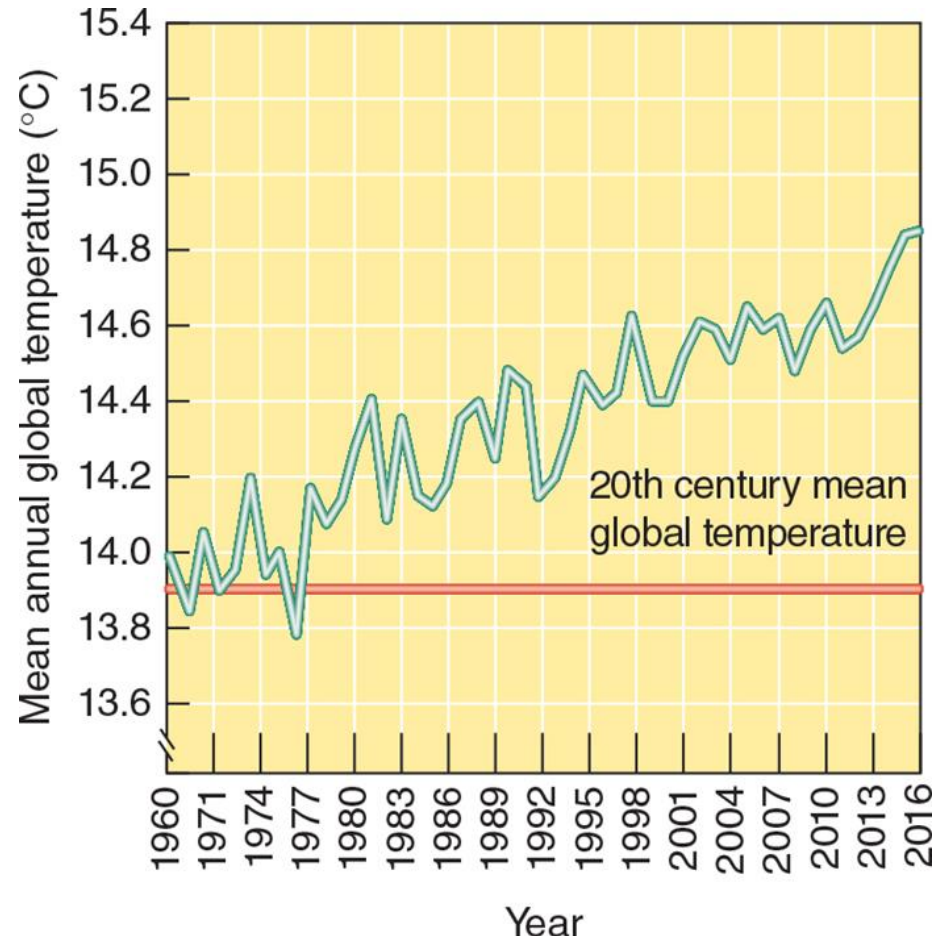
- Sea ice coverage expands in winter, contracts in summer
 - Can vary by 500,000km² annually
- Declining coverage
 - 1979 - coverage of 15.5million km²; 2017 - 13.5million km²
 - Last 10 years lowest coverage on record



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Introduction to Climate Change: Mean Annual Global Temperature 1960–2016

- For 2013-2016, each year set new average temperature record



Evidence for Climate Change

- 20 warmest years since mid-1800s have occurred since 1990
- Phenological spring in N. hemisphere now begins 6 days earlier (date the buds of specific plants open)
- U.S. has greater frequency of extreme heat-stress events
- Sea level rise - < 2 cm/decade for most of 20th century, now ~3 cm/decade
- Warming is not due to natural causes
 - Human produced greenhouse gases are most plausible explanation

Scientific and Governmental Agreement on Climate Change

- Intergovernmental Panel on Climate Change (IPCC) in 1988
 - International experts reviewed scientific literature and summarized evidence for climate change
- Emissions scenarios
 - Prediction about the amounts, rates, and mix of future greenhouse gases
- Predicted global average temperature increase of 0.2 degrees Celsius (0.4 degrees Fahrenheit) in each of the next 10 years

Climate Change Terminology (1 of 2)

- Greenhouse Gas
 - Gas that absorbs radiated heat from the sun, thereby increasing the temperature of the atmosphere
- Infrared Radiation
 - Radiation with wavelengths longer than visible light, but shorter than radio waves
 - Most energy absorbed by Earth is radiated as infrared, which greenhouse gases can absorb
- Radiative Forcing
 - Capacity of a gas to affect the balance of energy that enters and leaves the atmosphere

Climate Change Terminology (2 of 2)

- Parts Per Million (ppm)
 - A unit of concentration used for CO₂
- Aerosol Effect
 - Atmospheric cooling that occurs where and when aerosol pollution is the greatest

Greenhouse Gases (1 of 2)

Table 20.1 Increases in Selected Atmospheric Greenhouse Gases, Preindustrial Times to the Present

Gas	Estimated Preindustrial Concentration*	During post-industrial era Concentration
Carbon dioxide	288 ppm**	404 ppm
Methane	848 ppb***	1852 ppb
Nitrous oxide	285 ppb	328 ppb
Chlorofluorocarbon-12	0 ppt****	521 ppt
Chlorofluorocarbon-11	0 ppt	234 ppt

Greenhouse Gases (2 of 2)

*The preindustrial value is for the 17th and 18th centuries. There have been significant variations, as, for example, over the course of the ice ages.

**ppm = parts per million.

***ppb = parts per billion.

****ppt = parts per trillion.

Source: Carbon Dioxide Information Analysis Center, Environmental Sciences Division, Oak Ridge National Laboratory (historical estimates),

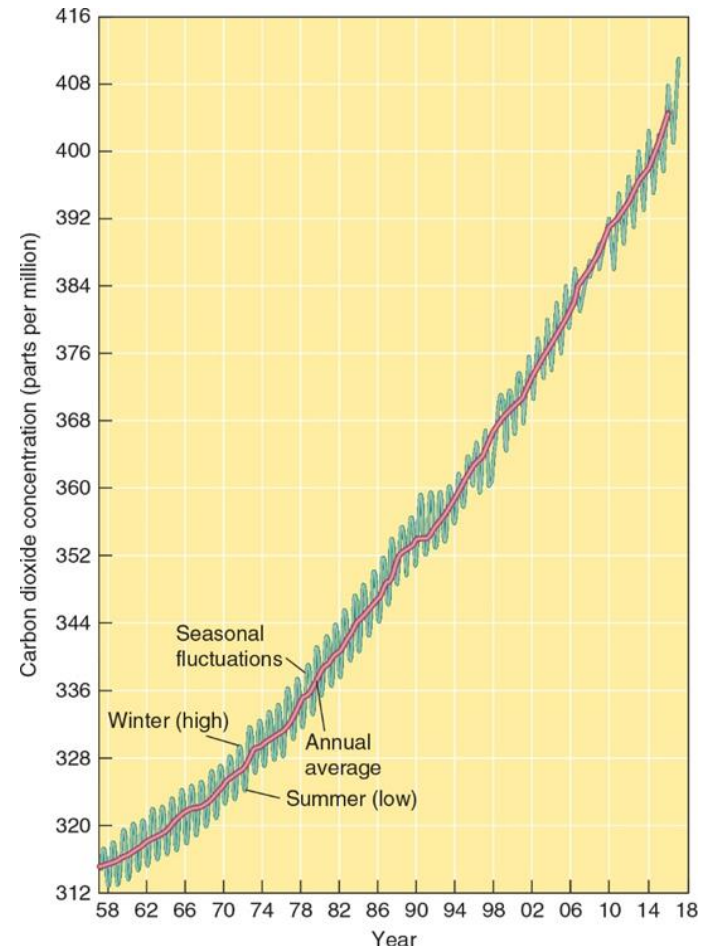
NOAA Annual Greenhouse Gas Index (2015 data).

Note that global chlorofluorocarbon production has decreased substantially since the 1970s, but chlorofluorocarbons are very stable and remain in the atmosphere for decades after being emitted.

- Greenhouse gas concentrations are increasing
- Increases of these gases and tropospheric ozone (O₃) are result of human activities

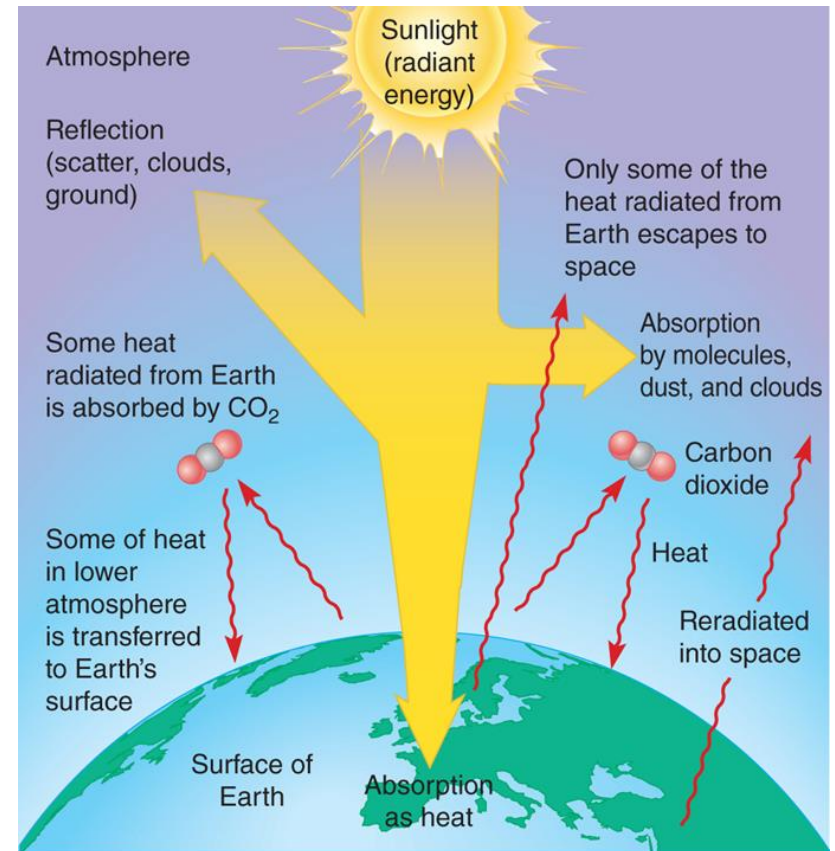
CO₂ in the Atmosphere, 1958-2017

- Steady increase in CO₂ since 1958
- Annual variation due to seasons
 - Plants actively growing in summer
 - Less activity in winter



Enhanced Greenhouse Effect

- Build up of greenhouse gases (CO_2 and others) absorbs outgoing radiation from Earth
- Warming atmosphere



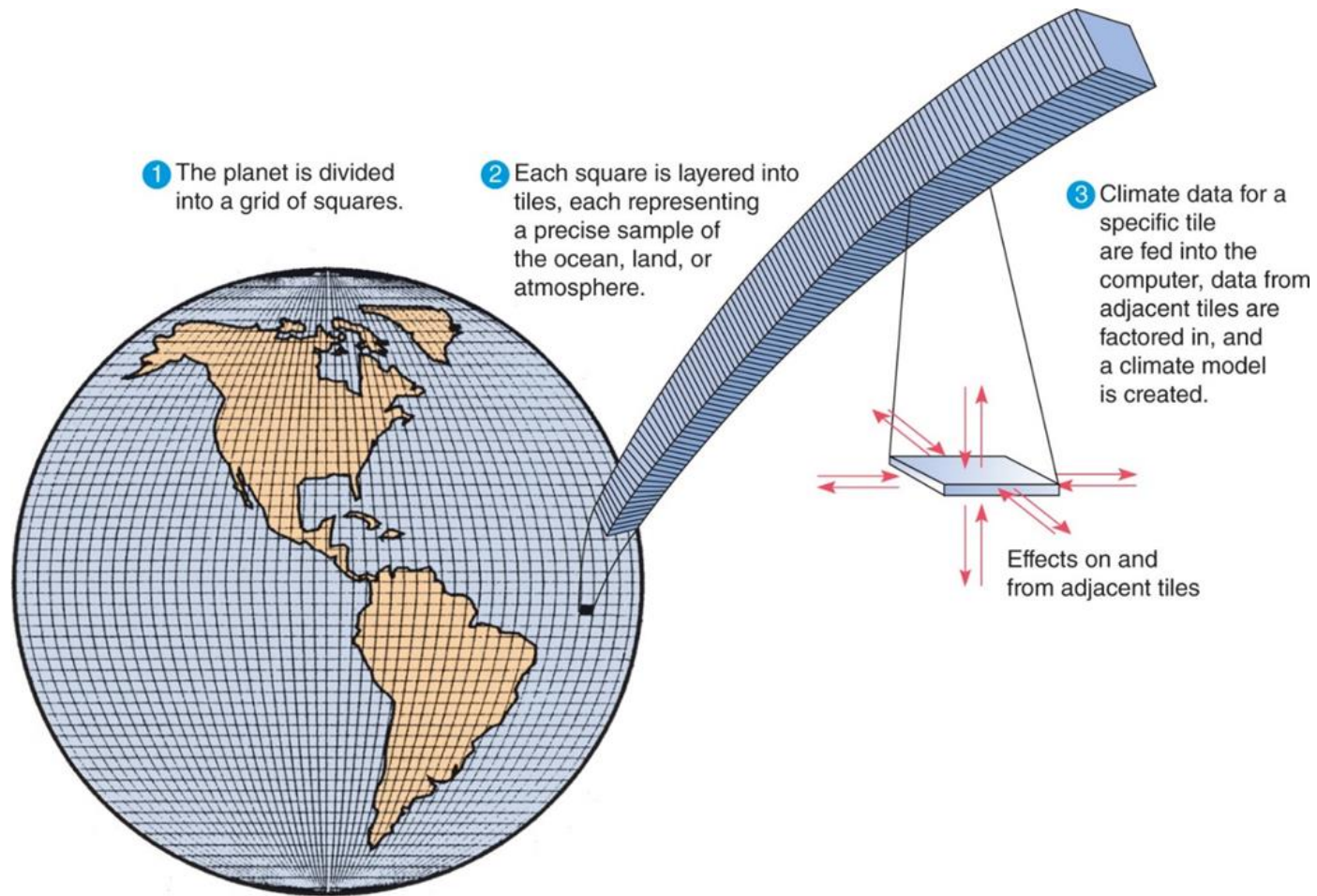
Pollutants That Cool the Atmosphere

- Atmospheric Aerosols
 - Both human and natural sources
 - Tiny particles (or sulfur) that remain in troposphere for weeks or months
 - Sulfur-laden layer in the atmosphere reduces the amount of sunlight reaching earth
- Complicates models of climate change
- Aerosol effect
 - Atmospheric cooling that occurs where and when aerosol pollution is the greatest

Climate Models

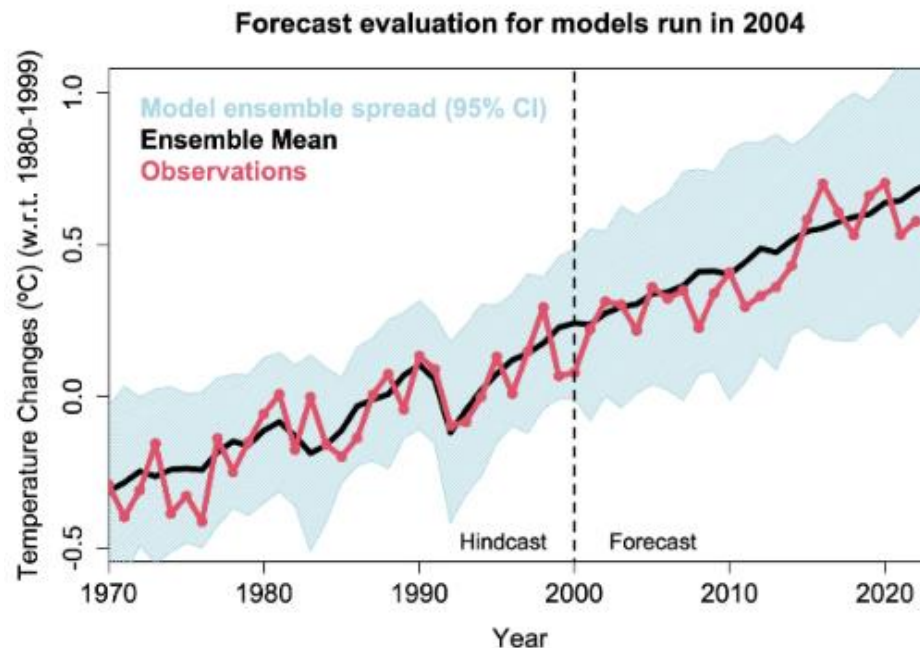
- Climate affected by:
 - Winds, clouds, ocean currents, and albedo
 - Albedo – measure of reflectivity (ice greater albedo than asphalt)
- Used to explore past climate events
- Advanced models can project future warming events
- Models are only as good as the data and physical laws used to program them
 - Limitations remain

Building a Climate Model



Study Confirms Climate Models are Getting Future Warming Projections Right (NASA)

Now a new evaluation of global climate models used to project Earth's future global average surface temperatures over the past half-century answers that question: **most of the models have been quite accurate.**



Climate Models (NASA)

Models that were used in the IPCC 4th Assessment Report can be evaluated by comparing their approximately 20-year predictions with what actually happened. In this figure, the multi-model ensemble and the average of all the models are plotted alongside the NASA Goddard Institute for Space Studies (GISS) Surface Temp Index (GISTEMP).

The GISTEMP v4 is an estimate of global surface temperature change. Graphs and tables are updated around the middle of every month using current data files from **NOAA GHCN v4** (meteorological stations) and **ERSST v5** (ocean areas).

Climate drivers were known for the 'hindcast' period (before 2000) and forecast for the period beyond. The temperatures are plotted with respect to a 1980-1999 baseline.

Credit: Gavin Schmidt Hansen et al. (2010) & Lenssen et al. (2019) <https://data.giss.nasa.gov/gistemp/>

Forecasting Global Warming

The findings of this analysis of earlier climate models increase scientists' confidence that both they and today's more advanced climate models are accurately forecasting global warming (Gavin Schmidt, director of NASA's Goddard Institute of Space Studies in New York).

This research might help clear up public misconceptions about the success of previous climate modeling efforts.

Climate Model Projections

NASA's Jet Propulsion Laboratory, By Alan Buis

To successfully match new observational data, climate model projections must encapsulate **climate physics** while also making accurate predictions about future **carbon dioxide emission levels** and other climate-influencing factors such as solar variability, volcanoes, and other human-produced and natural greenhouse gas and aerosol emissions.

This study's accounting for variations between **anticipated and actual emissions**, as well as other aspects, enabled a more concentrated review of the models' portrayal of the Earth's climate system.

<https://climate.nasa.gov/news/2943/study-confirms-climate-models-are-getting-future-warming-projections-right/>

DataSets: Goddard Institute for Space Studies



National Aeronautics and Space Administration
Goddard Institute for Space Studies

Goddard Space Flight Center
Sciences and Exploration Directorate
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ModelE Climate Simulations

Climate Simulations for 1880-2003

These webpages present data products related to the journal article "Climate simulations for 1880-2003 with GISS modelE"

For many climate diagnostic quantities are provided from the large number of climate paper. The available data include the climate responses to each of the forcings for a ocean C, Russell ocean) and a few model runs with a prescribed ocean (ocean A, flux ocean (ocean B, specified horizontal heat transport and vertical diffusion into deep se, the data are organized here along the lines of the tables 1 and 3 in that paper.

selecting "Plots" from one of the following groups.

and Responses	Plots	Table PDF
Table 3: Simpler Ocean Representations	Plots	Table PDF

(Note: PDF documents require a special viewer such as the free [Adobe Reader](#).)

In the tables and plots:

- "Responses" are obtained using a coupled atmosphere-ocean model with prescribed time varying forcings. Unless otherwise noted, the results are based on ensembles of 5 runs
- "Fi", the instantaneous forcing, is the radiative flux change at the tropopause after the forcing agent is introduced.
- "Fa", the adjusted forcing, is the flux change at the top of the atmosphere (and throughout the stratosphere) after the



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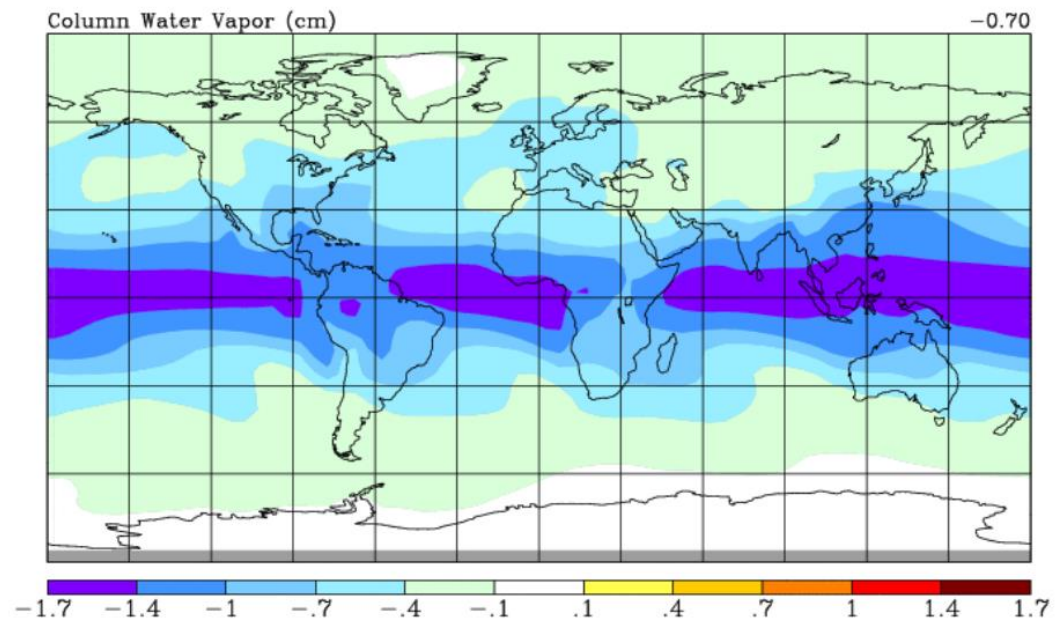
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Efficacy of Climate Forcings

Maps: 100-yr ANN Response to $1/8 \times \text{CO}_2$

Sources: E2CO2E - 100-yr ANN Response to $1/8 \times \text{CO}_2$





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"Efficacy of Climate Forcings" Simulations

Maps: 100-year Response to 1/8 x CO2

Quantity:

Column Water Vapor (cm)

Mean period:

Output:

Temperature

Ground Temperature, top layer (° C)
Surface Air Temperature, mean (° C)
Surface Air Temperature, daily maximum (° C)
Surface Air Temperature, daily range (° C)
MSU Channel 2 Temperature (° C)
MSU Channel 3 Temperature (° C)
MSU Channel 4 Temperature (° C)

Water Vapor

Column Water Vapor (cm)
Liquid Water path (.1 mm)
Precipitation (mm/day)
Evaporation (mm/day)

Runoff

Total Runoff/land (mm/day)
Ground Runoff/soil (mm/day)
Underground Runoff/soil (mm/day)
Runoff/glacial ice (mm/day)

Pressure

Sea Level Pressure (mb)

[Return to "Efficacy" Homepage](#)



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- "Fa", the adjusted forcing, is the flux change at the top of the atmosphere (and throughout the stratosphere) after the stratosphere is allowed to adjust radiatively to the presence of the forcing agent.
- "Fs", the fixed SST forcing, is a combination of the flux change at the top of (and throughout) the atmosphere and of the global surface air temperature change after the forcing and with observed sea surface temperature (SST) and sea ice (SI) held fixed.

See the [model forcing pages](#) for a more detailed description of these forcings.

Table 1. Forcings and Responses for Greenhouse Gas Experiments

	Forcing Mechanism	Fi(W/m ²)	Fa(W/m ²)	Fs(W/m ²)	Fixed SST Response	100-yr Response		
1	Scaled GHG-concentrations (1xCO ₂ = 291ppm)							
	1/8 × CO ₂	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Hgt	Lat-Lon	Lat-Hgt
	1/4 × CO ₂	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Hgt	Lat-Lon	Lat-Hgt
	1/2 × CO ₂	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Hgt	Lat-Lon	Lat-Hgt
	5/4 × CO ₂	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Hgt	Lat-Lon	Lat-Hgt
	3/2 × CO ₂	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Hgt	Lat-Lon	Lat-Hgt
	2 × CO ₂	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Hgt	Lat-Lon	Lat-Hgt
	4 × CO ₂	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Hgt	Lat-Lon	Lat-Hgt
	8 × CO ₂	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Hgt	Lat-Lon	Lat-Hgt
2	Specified GHG-changes							
	CO ₂ ,CH ₄ ,N ₂ O,CFC 1880->2000	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Hgt	Lat-Lon	Lat-Hgt
	CO ₂ 291->370 ppm	Lat-Lon	Lat-Lon	—	—	—	—	—
	CH ₄ 837->1752 ppb	Lat-Lon	Lat-Lon	—	—	—	—	—
	CH ₄ 837->3504 ppb	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Hgt	Lat-Lon	Lat-Hgt
	CH ₄ 837->9000 ppb	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Hgt	Lat-Lon	Lat-Hgt
	N ₂ O 278->316 ppb	Lat-Lon	Lat-Lon	—	—	—	—	—
	N ₂ O 278->1800 ppb	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Lon	Lat-Hgt	Lat-Lon	Lat-Hgt

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Datasets and Images

Datasets and derived material are available from the NASA GISS websites for the following research projects:

Climate Simulations

GISS ModelE

- Earth's Energy Imbalance
- Efficacy of Climate Forcings
- Climate Simulations for 1880-2003
- Dangerous Human-Made Interference
- Transient Simulations with Beryllium-10
- Simulated Greenland Surface Mass Balance
- Desert Dust, Industrialization and Agricultural Fires

SWING2 Stable Water Isotope Intercomparisons

Climate Forcings

- Forcings in GISS Climate Model
- Dust Mineral Fractions at Emission
- Globally Uniform Mineral Dust Aerosols
- Global Land Cover Datasets
- Surface Exchange of Carbon Dioxide
- Surface Exchange of Methane

Climate Impacts

- AgMIP Climate Forcing Datasets for Agricultural Modeling
- Evolving Fragility of the Global Food System
- Global Fire Weather Database (GFWED)

Earth Observations

Temperature

- GISS Surface Temperature Analysis (GISTEMP)
- GISTEMP Uncertainty Quantification
- Common Sense Climate Index (CSCI)

Aerosols

- Research Scanning Polarimeter (RSP)
- Global Aerosol Climatology Project (GACP)
- Stratospheric Aerosol and Gas Experiment II (SAGE II) V6.20

Clouds

- International Satellite Cloud Climatology Project (ISCCP)
- GEWEX Cloud System Study Data Integration for Model Evaluation (GCSS-DIME)
- Cumulus And Stratocumulus Cloudsat-CALipso Dataset (CASCCAD)

Oceans

- Global Seawater Oxygen-18 Database
- Ocean Carbon States — Database and Toolbox

Precipitation

- Observed Land Surface Precipitation Data: 1901-2000 (CRU TS 2.0)
- Observed Land Surface Precipitation Data: 1850-1995

<https://data.giss.nasa.gov/>

Modeling Environmental Complexity

- Environmental feedbacks are complicated
 - Positive or negative feedbacks
 - Type of cloud production can affect result: low-lying clouds – negative, high thin cirrus clouds - positive
- Most models project climate a few decades or a century from now
 - Based on different abatement strategies which will affect future CO₂ concentration

Unpredictable and Extreme Climate Change

- Projecting future climate with incomplete knowledge of environmental feedbacks
 - Solid scientific understanding, but environmental complexities abound
- abrupt climate change?
 - Occurrence of a large, rapid change in climate
 - Examples:
 - Ocean conveyor belt affected by melting of freshwater in Arctic
 - Portion of ice sheet breaking off
 - Polar vortex – cold air from poles move toward equator

Effects of Global Climate Change

- Wildfires in California
- Since 1985, wildfires are
 - More frequent
 - Lasting longer
 - Longer fire season
- Main drivers
 - Higher temp
 - Earlier snowmelt
 - Some land use
 - change

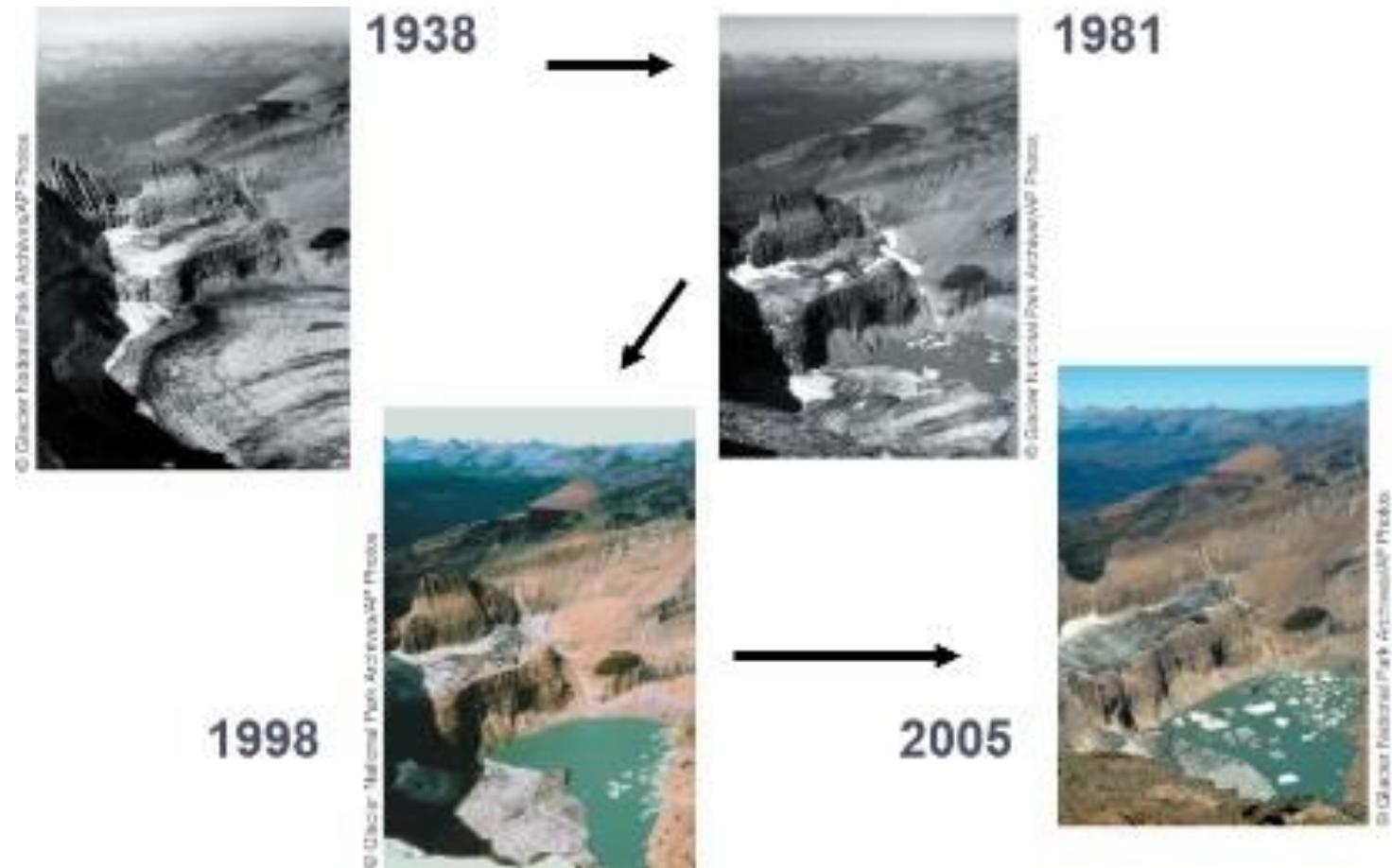


ROBYN BECK/AFP/Getty Images

Melting Ice and Rising Sea Levels

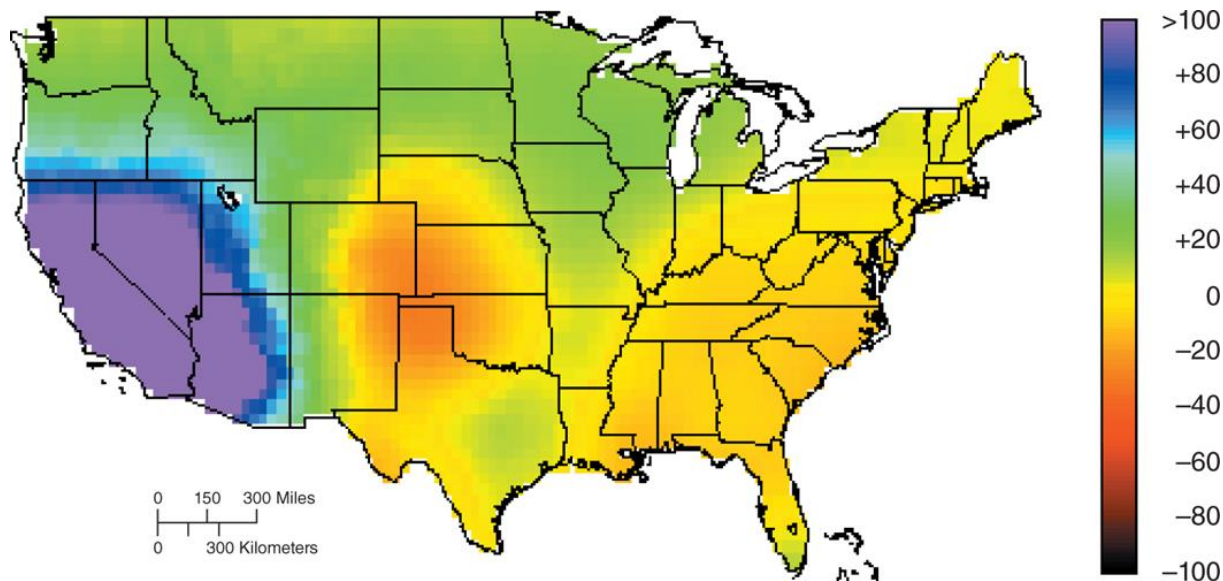
- IPCC projects sea-level rise of 18-59 cm by 2100
- Sea level rise caused in 2 ways:
 - Thermal Expansion
 - Melting of land ice
- Melting has positive feedback
 - Increased melting decreases ice, which decreases albedo leading to further warming

Grinnell Glacier, Glacier National Park



Changing Precipitation Patterns

- Some areas will get more water, some areas will have greater droughts
- Frequency and intensity of storms may increase



Effects on Organisms

- Zooplankton in parts of California Current have decreased by 80% since 1951
 - Affects entire food chain
- Species have shifted their geographic range
- Migrating birds are returning to summer homes earlier
- Ecosystems at greatest risk of species loss (short term):
 - coral reefs, mountain ecosystems, coastal wetlands, tundra, and polar areas

Zooplankton are small, aquatic microorganisms in the water column that include open water insect larvae and aquatic mites.

Effects on Coral Reefs

Coral reefs are massive structures made of limestone deposited by coral polyps (overgrowth in any organ)

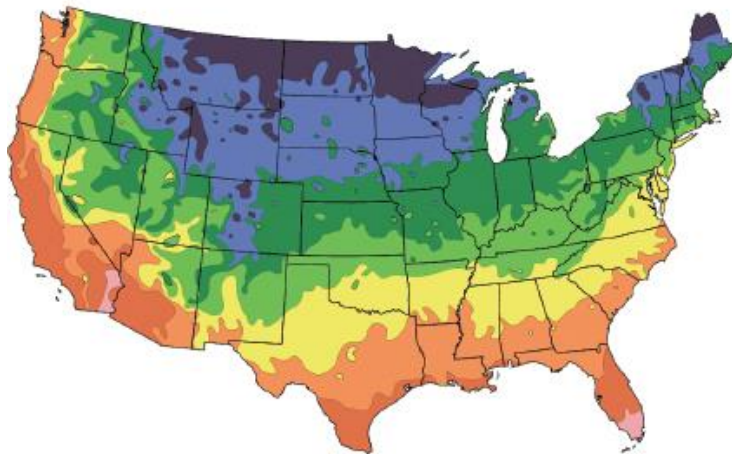
- Coral reefs can be bleached (right) due to increase in water temperature and acidification
 - Coral ejects symbiontes
 - Coral is more susceptible to stress and disease



Effects on Vegetation

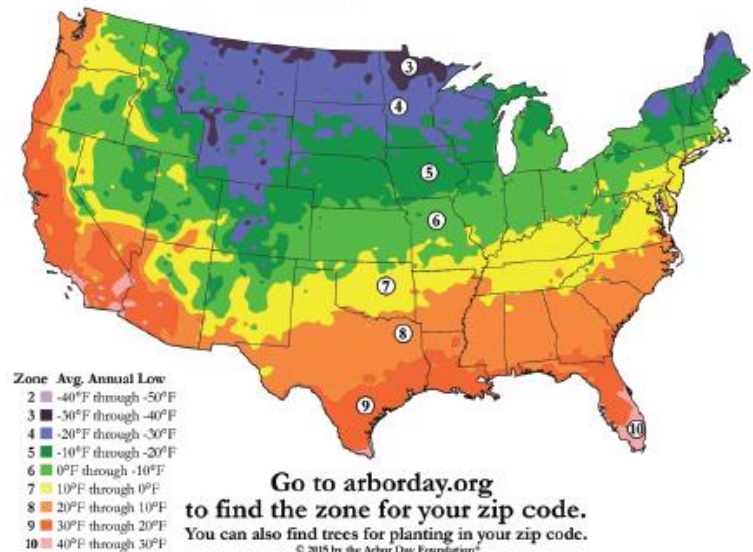
- Warming necessitated update to Plant Hardiness Zone Map in 2015
 - Change in plants best suited to location

1990 Map



After USDA Plant Hardiness Zone Map, USDA Miscellaneous
Publication No. 1475, Issued January 1990.

2015 arborday.org Hardiness Zones Map



Zone Avg. Annual Low
2 -40°F through -50°F
3 -30°F through -40°F
4 -20°F through -30°F
5 -10°F through -20°F
6 0°F through -10°F
7 10°F through 0°F
8 20°F through 10°F
9 30°F through 20°F
10 40°F through 30°F

Go to arborday.org
to find the zone for your zip code.
You can also find trees for planting in your zip code.
© 2015 by the Arbor Day Foundation®

Effects on Human Health (1 of 2)

Table 20.2 Some Known and Expected Effects of Climate Change on Human Diseases

Type of Disease	Known Effects	Predicted Effects
Vector-borne* disease (e.g., Lyme disease, malaria)	Warming-related expansion or shift of geographic range of disease vector. Precipitation-related expansion or shift of geographic range. Travel-related risk of disease. Changes in temperature, precipitation could lead to major ecosystem shifts, impacting disease burden and incidence.	Increased transmission of some diseases transmitted from animals to humans. Establishment of novel imported infectious diseases. Regional variability in changes; some areas will be impacted more than others.
Water- and food-borne disease (e.g., E. coli, Salmonella)	Temperature-related increase in survival of disease organisms. Precipitation-related shifts in geographic range of disease organisms. Increased temperature expected to shift northernmost tolerance of many diseases.	Increased intensity and frequency of water- and food-borne diseases, particularly in the northernmost parts of the United States and Canada.

Effects on Human Health (2 of 2)

Type of Disease	Known Effects	Predicted Effects
Respiratory disease (e.g., influenza, Streptococcus bacillus)	Shorter/warmer winter could reduce disease incidence and burden. Migration of nonimmune populations to areas with endemic disease increases incidence. Air pollution changes could cause/exacerbate disease.	Reduced number of respiratory diseases with cold weather.** Increased respiratory diseases associated with dense populations.** Increased respiratory diseases associated with diminished air quality.**
Fungal diseases (e.g., Blastomyces dermatitis, Cryptococcus gatti)	Soil ecology changes could shift range of fungus populations. Warm, dry summers followed by warm, wet winters could increase fungus counts. Changes in temperature, precipitation, and humidity could expand geographic range of some fungi.	Growing conditions that favor infectious fungal spores. Regional variability in changes of numbers and types of fungi.

*Vectors are the animals that carry a disease organism or a stage of a disease organism. Mosquitoes are a common vector, but many diseases require a mammalian vector (e.g., fox, deer, rat).

**It is not clear which of these effects will dominate.

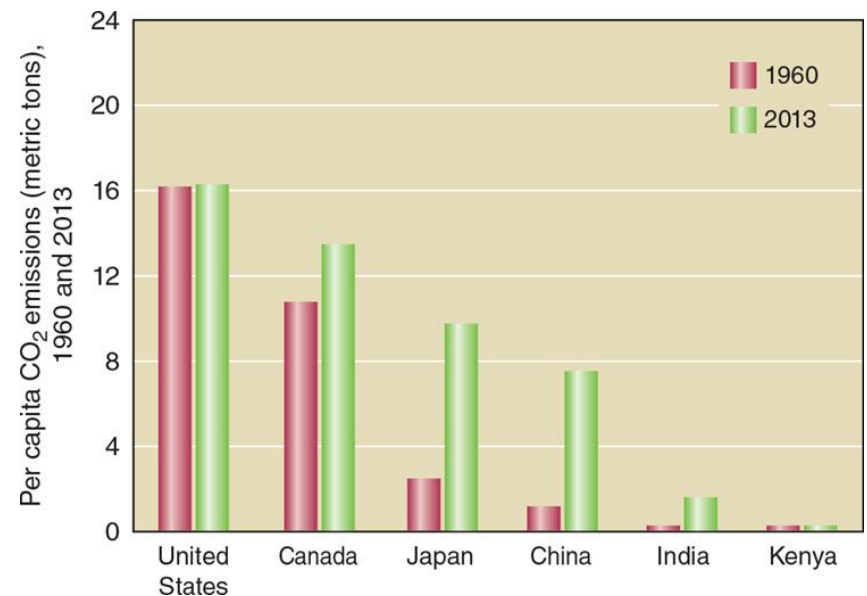
Source: Greer, Ny, and Fisman. "Climate Change and Infectious Diseases in North America: The Road Ahead." Canadian Medical Association Journal (2008).

Effects on Agriculture

- Difficult to anticipate
 - Productivity will increase in some areas and decrease in others
- Rise in sea level may inundate flood plains and river valleys (lush farmland)
- Changing nighttime temperatures
- Increase in drought frequency and duration
- Effect on pests is unknown
- Location (i.e., elevation and altitude) where certain crops can be grown will change

International Implications of Climate Change

- Developed and developing countries interest and ability to decrease climate change
 - Differing self-interests
 - Differing ability to meet the challenges of climate change



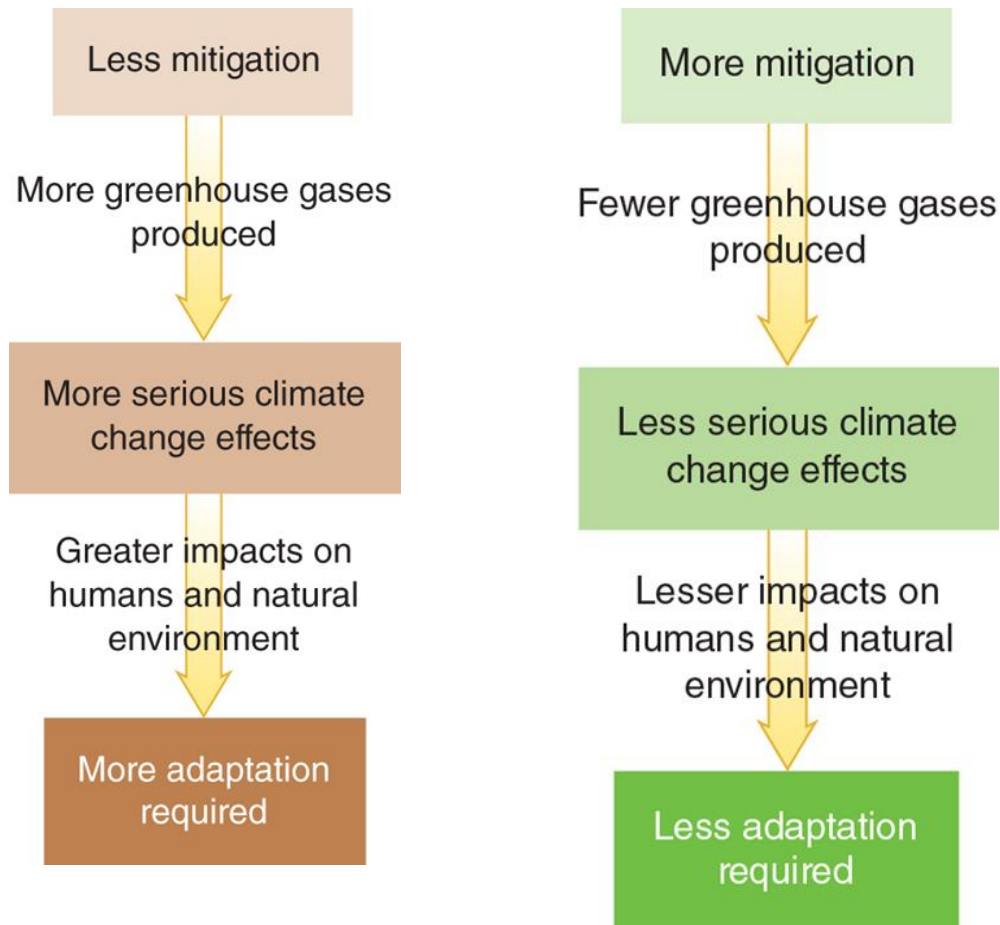
Links with Ozone Depletion and Acid Deposition

- Combined effects may increase ecosystem damage
 - May increase evaporation from water bodies
 - Evaporation can concentrate dissolved compounds in water
 - UV radiation may penetrate deeper into water
 - Disrupt photosynthesis
 - Cause sunburn damage

Dealing with Global Climate Change

- Two ways to manage climate change:
 - Mitigation: Limiting greenhouse gas emissions to slow down or postpone global climate change
 - Adaptation: Developing behaviors and technologies that allow humans to live with the environmental changes and societal consequences brought about by global climate change

Relationship Between Mitigation and Adaptation



Mitigating Global Climate Change

- Locate/invent alternative fuels to fossil fuels
- Increase efficiency of cars and trucks
- Energy-pricing strategies
- Carbon capture and storage
- Naturally sequester carbon in trees
- Geoengineering
 - Global scale projects to mitigate effects
 - Example: seeding ocean with iron to stimulate phytoplankton growth and CO₂ storage in deep ocean

Adapting to Global Climate Change

- Rising sea levels may displace coastal populations
 - Move inland
 - Construct dikes and levees
- Adapt to shifting agricultural zones
- Update city infrastructure – NYC sewer line
- Climate change refugees
 - People leaving homes because of climate change effects
 - Example: rising sea levels, storm surges

International Efforts to Reduce Greenhouse Gas Emissions

- UN Framework Convention on Climate Change (UNFCCC)
- In 2015, met in Paris where 132 countries made a binding agreement toward mitigation
 - China, India, Mexico, and Canada; the United States has not committed.
- Three goals of agreement:
 - Keep global avg. temperature increase below 2 degrees Celsius
 - Improve capacity to adapt
 - Enable effective financing